ADAPTING SUSTAINABLE DESIGN PRINCIPLES FOR THE USE OF THE PRODUCT DESIGN PROCESS

Lappeenranta–Lahti University of Technology LUT

Masters's Programme in Mechanical Engineering, Master's thesis

2025

Pekko Räsänen

Examiners: Professor Juha Varis

Associate professor Amir Togiani

ABSTRACT

Lappeenranta–Lahti University of Technology LUT

LUT School of Energy Systems

Mechanical Engineering

Pekko Räsänen

Adapting sustainable design principles for the use of the product design process

Master's thesis

2025

70 pages, 2 figures, and 3 appendices

Examiners: Professor Juha Varis, Associate professor Amir Togiani

Keywords: sustainability, sustainable design, sustainable design principles

Sustainability as topic has increased its importance in last decades. Because this many businesses need to consider sustainability of their operations. Part of these considerations is the product development and the processes within it, in which the so called "Sustainable Design Principles" play a role to help to guide the work of the product design teams to create more sustainable products. In this thesis such principles were created for the use of the company that commissioned the work. The methodology used in this work includes literature review, review of the industry practices and interviews with selected group of employees of the company. Based on the findings of these methods, collection of seven Sustainable Desing Principles was proposed.

TIIVISTELMÄ

Lappeenrannan–Lahden teknillinen yliopisto LUT

LUT energiajärjestelmien tiedekunta

Konetekniikka

Pekko Räsänen

Kestävän suunnittelun periaatteiden soveltaminen tuotekehitysprosessiin

Konetekniikan diplomityö

2025

70 sivua, 2 kuvaa, ja 3 liitettä

Tarkastajat: Professori Juha Varis, Tutkijaopettaja Amir Togiani

Avainsanat: kestävä kehitys, kestävä suunnittelu, kestävän suunnittelun periaatteet

Kestävä kehitys on aihealue, jonka merkitys on lisääntynyt viime vuosikymmeninä. Tämän takia yhä useamman yrityksen on huomioitava aihe toiminnassaan. Osan toimintaa muodostaa tuotekehitys ja sen prosessit, joissa niin kutsutut kestävän suunnittelun periaatteet auttavat ohjaamaan tuotesuunnitteluosastoja kestävän kehityksen arvojen mukaisten tuotteiden suunnittelussa. Tässä diplomityössä määritetään kyseiset periaatteet tehtävän antaneen yrityksen käyttöön. Metodeina käytetään kirjallisuuskatsausta, alan käytäntöjen tarkastelua sekä haastatteluja aiheen antaneen yrityksen työntekijöille. Kootun tiedon perusteella ehdotetaan seitsemää erilaista kestävän suunnittelun periaatetta.

Table of contents

Abstract

Tiivistelmä

Declarations

1 Introduction 7
1.1 Short introduction to sustainability 8
1.2 Background of the study 9
1.3 Introduction of Halton Group 10
1.3.1 Halton Foodservice 11
1.3.2 Halton Buildings 12
1.3.3 Halton MEI 12
1.3.4 Halton Health and Cleanroom 13
1.3.5 Product examples 13
1.4 Research problem 15
1.5 Research questions 15
1.6 Objective of the study 16
1.7 Scope of the study 16
2 Methodology 17
2.1 Literature review methodology 17
2.2 Literature review 18
2.3 Review of practices in industry, goal and methodology 22
2.4 Interviews, their goal and methodology 23
3 Results 25
3.1 Results of review of current industry practices 25
3.2 Results of interviews 28
4 Discussion 35
5 Conclusions 46
6 Compendium 51
References .....

Appendices

Appendix 1. List of the reviewed sustainable design guidelines

Appendix 2. Examples of Halton products

Appendix 3. Topic guides for the interview

Figures

Figure 1. Halton Group organization

Tables

Table 1. Sustainability themes depending on where they appeared in research

DECLARATIONS

Turnitin

The originality of this thesis has been reviewed with the Turnitin similarity checking service.

AI usage

The author of this thesis, Pekko Räsänen, did not use any AI tools during the preparation of the thesis**.**

Responsibility

The author, Pekko Räsänen, takes full responsibility for the content of this thesis.

1 Introduction

Even though sustainability, or concepts around it have been surfacing to the discussion since the mid-20th century and even before, it came to the public limelight in the 1980's when the United Nations Brundtland commission defined term (1987). At the beginning it was merely topic of general discussion of the challenges laying ahead in the future (Barrero et al., 2021), but after the millennium its importance has increased, as limited nature of Earth's resources has become better known by the wider public (Larrinaga and Bebbington, 2021). In some cases, due increased knowledge, but in others due to sustainability related crises. Due increased interest even smaller and midsize companies are moving towards concrete applications to improve sustainability of their operations. First it formed good sales argument for the environmentally conscious customers, but today this is already an important requirement in supply chain, as many large companies are requiring sustainability performance from their suppliers. (Grunwald et al., 2025.) Requirements from the legislative side can also be expected to increase as the time goes on, while quickly changing and advancing requirements make it hard to remain up to date on incoming regulations. (Hummel and Jobst, 2024.)

Because of these expectations, it is beneficial for businesses to prepare their product development pipeline for these current and future needs. This applies also to Halton Group which as manufacturer of components of HVAC systems with a global representation faces expectations from all previously mentioned directions. For this purpose, Halton has been taking actions to make sustainability improvements. As part of these actions, set of sustainable design principles for the use of sales, operations and product design teams are to be drafted. The purpose of this work is to create that list of design principles best suited for the use of the Halton Group, based on the literature review, review of the industry practices, and the interviews performed inside the company. While this thesis is based on the needs of single company, and such is not universally applicable, it gives good idea of considerations needed when performing equal task for a similar organization.

1.1 Short introduction to sustainability

Sustainability as an expression needs to be defined as it may not be clear to everyone what elements are included in it (Moore et al., 2017). Previously mentioned Brundtland commissions definition – more than three decades later still stands as a benchmark for the definition of sustainability and is referred in many works. They (Brundtland commission, 1987) defined it in their report "Our Common Future" as follows; *"development that meets the needs of the present without compromising the ability of future generations to meet their own needs".* This single sentence definition contains all the basics of sustainability in easily understandable form (Santillo, 2007). To define the concept better the International Institute for Sustainable Development (1992) proposed more detailed definition, better suited for the commercial world; *"For the business enterprise, sustainable development means adopting business strategies and activities that meet the needs of the enterprise and its stakeholders today while protecting, sustaining and enhancing the human and natural resources that will be needed in the future"*. This definition gives elements that offer more direct guidance regarding actions needed to improve sustainability performance of a company.

This is a balancing act from the business point of view (Noeth, 2024). To exist a company must create profit and in manufacturing industries way to do this is to create a physical product. The goal is then to improve means of production and product design in such a way that negative effects can be minimized or compensated other ways (Ashby, 2012, 15-25).

For a more detailed view into the corporate sustainability, United Nations Global Compact has listed "*The Ten Principles of the UN Global Compact*"

> ***"Human Rights***
>
> *Principle 1: Businesses should support and respect the protection of internationally proclaimed human rights; and*
>
> *Principle 2: make sure that they are not complicit in human rights abuses.*
>
> ***Labour***
>
> *Principle 3: Businesses should uphold the freedom of association and the effective recognition of the right to collective bargaining;*

Principle 4: the elimination of all forms of forced and compulsory labour;

Principle 5: the effective abolition of child labour; and

Principle 6: the elimination of discrimination in respect of employment and occupation.

Environment

Principle 7: Businesses should support a precautionary approach to environmental challenges;

Principle 8: undertake initiatives to promote greater environmental responsibility; and

Principle 9: encourage the development and diffusion of environmentally friendly technologies.

Anti-Corruption

Principle 10: Businesses should work against corruption in all its forms, including extortion and bribery. While far from being only method of classification of sustainability indicators it is simple and easy to understand."

While these principles helped to change focus from the companies' facilities and product footprints to a broader view of sustainability as a large array of social challenges (Noeth, 2024), in this work most of the interest is concentrated on principles 7, 8 and 9 Especially principle 9 is relevant to the topic of product development.

1.2 Background of the study

Previously discussed needs for improved sustainability have also been faced by the Halton Group, a manufacturer of HVAC systems and components. Pressure for this has come from customer wishes, legislation as well as the company values and strategy; will to improve sustainability and promote its goal of enabling well-being in demanding indoor environments. (Halton Group, n.d.a) It is understood to be an essential part of business in

the future as sustainability requirements are only expected to grow (Tsalis et al., 2020, 1617-1621; Spalding et al., 2021, 52-55).

To answer this need, Halton Group has already started to follow the UN Sustainable Development Goals and has committed to the Science Based Targets Initiative (Halton Group, n.d.a) which is "*a global body established in 2015 that enables companies and financial institutions to set ambitious emissions reductions targets in line with the latest climate science. It is a collaboration between CDP, the United Nations Global Compact, World Resources Institute and WWF."* (WWF, 2025.) As part of its action to improve sustainability of its operations the Halton Group is developing a set of "Sustainable Design Principles" for the use of product design teams in all its SBA's – Strategic Business Areas. This master's thesis purpose is to help in forming suitable design principles. Although the aspect of product design is the most important viewpoint of this work, it must not be forgotten that these principles help understand goals and targets of Halton Group. That way they can be also used to improve co-operation between different departments inside Halton.

1.3 Introduction of Halton Group

To give better understanding of the Halton Group and its organisation and such limitations placed upon this work in this chapter a short introduction of company is given. Introduction is concentrated on the main organizational framework. Additionally, a short introduction to the main product groups and products of each business area is provided.

Halton was established in 1969 in Finland, originally the main products were store furniture, but already in the 70's future was seen in indoor air products. Halton continued to grow through 80's, and in 90's everything but HVAC related business was sold. Today Halton employs almost 2000 personnel and is present in over 35 countries with 316 million euros turnover (Halton, n.d.b; Lähteenmäki and Puustinen, 2019.)

Organisationally the Halton Group is divided into four different SBA's – Strategic Business Areas depending on type of customer they are concentrating on. These are Halton Foodservice, Halton Buildings, Halton MEI (Marine, Energy, and Infrastructure (+Service)) and fourth the newest part of the Halton, Health and Cleanroom. Each SBA also has internal organization concentrated on service business. Following graph will show basic

organizational structure of the Halton Group, its SBA's, operating locations of SBA's and its parent company Halton Ventures.

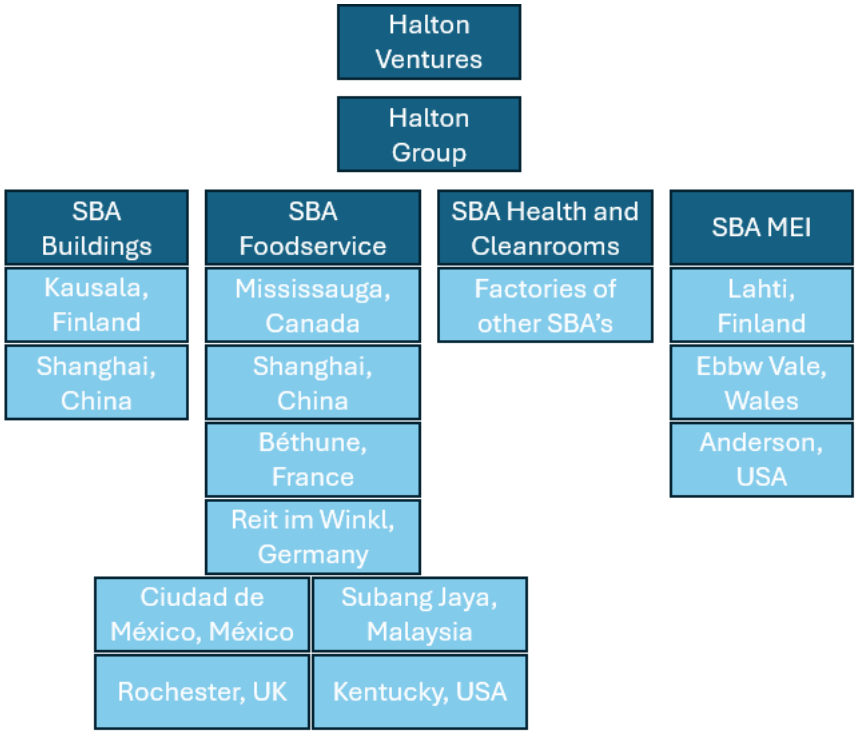


Figure 1. Halton Group organization (Source: author, based on data from Halton Group)

As can be seen from the graph each SBA manufacturers their products in multiple locations and countries. Many of these SBA's have their own products in which is the reason Halton's product range is large. (Halton, n.d.b.)

In following subchapter's, a basic introduction to typical products of each SBA is given. After introduction of SBA's there are few typical examples of product types to give an idea of typical construction and material.

1.3.1 Halton Foodservice

SBA Foodservice provides commercial kitchens and restaurants with indoor climate solutions. Main products consist commercial kitchen exhaust hoods, ventilated ceilings and

air handling units as well as their control systems, including demand-based systems. The main construction material is stainless steel. (Halton, Foodservice solutions, n.d.c.) Foodservice has factories in Mississauga – Canada, Shanghai – China, Béthune – France, Reit im Winkl – Germany, Subang Jaya – Malaysia, Ciudad de México – México, Rochester – United Kingdom, Kentucky – USA.

1.3.2 Halton Buildings

SBA Buildings concentrates on indoor climate components for public and commercial buildings, for example hotels and offices. SBA Buildings has wide selection of products, most important are dampers (fire, shut-off and flow control purposes), chilled beams, diffusers, grilles and plenums, Main materials used in the products are galvanized- and stainless steel and aluminium. (Halton, n.d.c) SBA Buildings has factories in Kausala – Finland and in Shanghai – China.

1.3.3 Halton MEI

SBA MEI contains three distinct customer groups. Marine sells products to ships, including passenger vessels, navy and other government vessels. Energy deals with offshore energy projects including oil and gas drilling rigs as well as offshore wind projects. Industry handles the large industrial projects including but not limited to nuclear-, hydrogen- and chemical plants.

Main products of the MEI are marine- and CE-certified fire dampers, gas-tight flow control and shut-off dampers, droplet separators, cabin ventilation units and kitchen exhaust hoods specially designed for shipborne use. Basic material of products is galvanized- and stainless steel. (Halton, n.d.c) MEI has factories in Lahti – Finland, Shanghai – China, Ebbw vale – Wales and in Anderson – USA.

1.3.4 Halton Health and Cleanroom

SBA Health and Cleanroom is the newest of group, formed by separating it from the SBA Buildings. This SBA provides solutions for medical operating rooms and the pharmaceutical industry that requires clean rooms or laboratories. (Halton, n.d.c.) Products of Health and Cleanrooms are combined from the offerings of other SBA's or sourced from suppliers depending on the project (Saxen, 2025). Health and Cleanroom products are made in the other Halton factories, mainly in Crépy and Kausala. While some components are from the external suppliers.

1.3.5 Product examples

This chapter gives typical examples of previously mentioned products, their materials and manufacturing methods. Appendix 2 contains pictures of the mentioned products.

Kitchen exhaust hood

Kitchen exhaust hoods are mounted in the ceiling with purpose of extracting fumes, steam and condensate from the kitchen environment. Hoods vary in size and design depending on the installation location. The simplest models are just designed to capture steam and efficiently remove it from the room. Exhaust hoods installed over cooking stations include different solutions to reduce grease buildup to ducting and equipment to automatically wash filters of unit. The construction material of a kitchen hood is almost exclusively stainless-steel sheets cut into shape with a punch and laser combination machines, shaped with press brakes and welded. Electrical components, actuators, lights, uv-tubes and junction boxes are sourced from external suppliers. Galley exhaust hoods designed for marine use are mainly similar with hoods designed for land-based kitchens (Halton, n.d.c).

Fire- and flow control dampers

Building codes require all openings in fire-rated walls to have fire dampers that automatically close in case of fire. Different types of fire dampers are manufactured depending on the type of duct and location they are installed. Fire dampers are mostly constructed from galvanized or stainless steel using similar manufacturing methods as kitchen hoods, cutting, bending and welding. Externally sourced parts include electrical or pneumatic actuators, switches and junction boxes.

Marine fire dampers are very similar to models designed for building installation, but typically the material of the frame is thicker according to SOLAS requirements and marine models are commonly multiple blade type, ie. duct opening is closed with multiple narrow blades instead of one large plate.

Flow control dampers are very similar with fire dampers from a manufacturing point of view and in multiple models share basic frame and blade construction. However, they lack testing and insulation of fire dampers and may contain extra equipment to measure and control airflow. (Halton, n.d.c.)

Droplet separators

Droplet separators are designed to reduce amount of water particles carried by the airflow to ducting of building/vessel. The construction of the frame is like one in dampers, but instead of damper blades, they feature separator vanes in airflow that capture droplets. The materials are stainless steel and aluminium. (Halton, n.d.c.)

Cabin Ventilation Units

Cabin ventilation units are used to control airflow through diffusers to indoor spaces as well as heat the incoming air if needed. The material of construction is mostly galvanized steel, and the manufacturing methods are similar to those of other products. (Halton, n.d.c.)

Diffusers, grilles and plenums

Diffusers, grilles and plenums are designed to distribute air into indoor spaces. They are manufactured from stainless and galvanized steel, aluminium and sometimes plastic.

Chilled beams

Chilled beams are ventilation systems installed in ceilings to provide cooling and heating. They are manufactured from a combination of galvanized steel, copper and aluminium in mixed construction.

1.4 Research problem

Sustainability goals and performance targets for design are typically listed at general level, but rarely are instructions given on how to apply them to actual product design (Blizzard and Klotz, 2012, 456-459). If these general sustainability goals and targets are directly passed to the product designers, the results may be less than optimal, especially in larger organizations as this will result multiple people determining their own methods of applying goals and targets (Baldassarre et al., 2024, 40-41). While in some cases this may work, usually this will result in incoherent approach over the organization, which will ultimately yield suboptimal results and such systematic approach is needed.

1.5 Research questions

To guide research, research questions are needed. Different research questions were formed and reviewed for their suitability. Three research questions were selected to be used. They are as follows:

- What kind of guidelines industry typically use for sustainable design?
- What are the possibilities and challenges related to sustainable design principles in the Halton Group?
- What concrete principles/guidelines should be followed in product design process to create sustainable HVAC system components?

First one was selected to see what kind of practices industry has formed so far.

Every company has their own set of possibilities and challenges. To create sustainable design principles interviews will be conducted inside company to find Halton specific ones. Unless this is done resulting set of principles is just general estimate instead of tailored set.

Final question will directly help to answer to the research problem and fulfil the objective of this study.

1.6 Objective of the study

Objective of this study is to create a set of principles that will guide Halton Group's product design teams in their work to create sustainable products while keeping their functionality at necessary level. As well as help other Group's functions in understanding sustainable considerations in products.

1.7 Scope of the study

The first limitation of the project scope is the Halton Group's requirement for principles to be suitable for group-wide application for the use of product design teams in all SBAs previously introduced. As these operating locations are in 10 different countries and such have their own local challenges and product groups the depth of this work remains general level. Specific details or ideas how to apply design principles are not included unless used as form of examples.

Additionally, the following more detailed limitations to the study scope were added. Suitability is limited only to Halton Group's main products/product categories, atypical products / one offs were left outside of consideration. This decision was made to reduce the change that single product affects design principles much more than its true share of production and sales.

This work concentrates on sustainability elements that apply to product design from the viewpoint of the product design team. It does not consider the social side of sustainability unless it is relevant to product design or the design work itself.

2 Methodology

This chapter will include a literature review performed at the start of research to find out current views on topic in scientific literature and to help to determine topics to be discussed in interviews and to help categorize the information found during the review of industry practices. Chapter also introduces the methods selected for the work; review of industry practices concerning sustainable design and interviews conducted within the Halton Group and discusses suitability of selected methods for their purpose. This source triangulation improves reliability of research (Tuomi and Sarajärvi, 2018).

2.1 Literature review methodology

Review of publications related to sustainable design was performed to find out current view on topic and what general methods of sustainable design are recognized and recommended. The literature review was performed as a general conceptual literature review. The purpose of literature review in this research is to gain general idea of the topic, a systematic review was seen too extensive.

A literature review was performed in two stages; in first suitable material searched was selected, and in second stage it was reviewed and summarized under reoccurring themes. Publications were searched from LUT University library's LUT Primo database. The initial search was performed using keywords: sustainab* AND design OR product, after initial search, search was extended with found keywords to include; design AND efficiency OR one of following keywords; product service systems, repairability, re-manufacturability, remanufacturing, design for disassembly, recyclability, product stewardship, modularity, disassembly, dematerialization, refurbishment. In some cases, common abbreviations of the previously mentioned keywords were used to extend the reach.

In selection process publications were read and their relevancy to topic was considered. Publications not relevant to the researched topic were discarded. In the selection process well-cited peer-reviewed publications were preferred. As new sources as possible were selected, in some cases even new publications referred to the older works, in these cases

older references were accepted. Additional searches were performed as needed to remove the need for secondary referencing.

2.2 Literature review

A literature review was performed as described in the previous chapter. From publications reviewed topics that appeared commonly in the referred material and were deemed most relevant for this work were selected. The topics are presented in alphabetical order based on the names they are commonly known. For each topic, a basic description is included with information why it is considered important in field of sustainable design. On some occasions elements in different topics overlap, this can be seen especially with modularity featuring as element in multiple topics.

(Desing for) Dematerialization

Dematerialization can be simply explained as a more effective use of inputs (Kemp-Benedict, 2018, 212). There is clear benefit when same product performance can be achieved with fewer raw materials. Additionally, there is benefit at the end of life of the product. While metals can be reused almost without limits, other raw materials will suffer loss of quality and functionality even in a well performed recycling process. (Smil, 2013, 119.) With the limited nature of the world's resources, for this reason trusting just the recyclability of a product is not enough. With dematerialization this loss can be reduced.

While the reduction of volume is the most common way to dematerialization of a product it should not be always seen as the only method of doing so. In some cases, materials or parts of product can be substituted with a completely different solution. The change from cathode-ray tube screens, which contain heavy metals like lead to modern LCD and LED technology can also be seen as an example of dematerialization (Kasulaitis et al., 2021, 436.)

(Desing for) Disassembly

According to Go et al. (2015) disassembly is the most important factor when determining how efficiently a product can be reused or remanufactured, otherwise important features considering sustainability. It is the first stage of discarding process, and it affects the level at which materials or components can be reclaimed from end-of-life products. As

disassembly can be considered process that does not add value, the time spent to it should be minimized.

Bogue (2007) lists three disciplines as essential for the proper application of design for disassembly:

1. the selection and use of materials;

2. the design of components and the product architecture;

3. the selection and use of joints, connectors and fasteners

(Desing for) Efficiency

Unlike some other design strategies, the benefits of a products energy efficiency are clear and wished also by customers (Faiers et al. 2007). An energy efficient product performs its task with the minimum possible energy consumption. The ISO 50001 standard defines it as follows: "*ratio or other quantitative relationship between an output of performance, service, goods, commodities, or energy, and an input of energy* " (SFS-EN ISO 50001:2018). According to Li et al. (2019) the energy efficiency of product is mostly determined by decisions made during initial design process of product. It needs to be recognized by design teams that these early decisions may limit options in latter part of product development or choices in rest of system. They mention that several studies have been conducted to determine energy consumption of different products which help for rough estimation of energy consumption, but more detailed design guides are still missing, instructions remaining at a general level.

(Desing for) Modularity

The ability to adapt components or products to different needs through modularity is seen as an important way to improve product sustainability. Benefits of modularity can be seen through whole lifecycle of a product. At the manufacturing stage it allows manufacturers easily adapt to the needs of the market, improving quality while reducing costs and lead times. As can be seen most of these benefits are for the manufacturer. During the use of the product sustainability benefits are seen when modularity allows easier repairability or upgradeability to extend the product lifespan. This also benefits the customer. (Sonego et al., 2018 199-201) According to Go et al., (2015, 23), during end of life of a product modularity improves sustainability of design by enhancing effect of other sustainability solutions,

disassembly for recycling or disassembly. If a product is designed to be remanufactured modularity may provide benefits also in that. Overall, modularity can be seen as important feature in the sustainable product design, but only if designers understand how it should be applied (Sonego et al., 2018, 199-201).

(Desing for) Product Service Systems (PSS)

Product service systems provide unique possibilities to commercial businesses in form of instead of selling products to customers (typically as low cost as possible) giving, at least partial control of product throughout its lifecycle. In typical PSS the customer purchases "solution" instead of a product, ownership and its risks remain with the company providing the said solution. From the customer point of view the benefit is avoidance of these risks of ownership, while for the supplier benefits from increased control over the product. Sustainability benefits can be realized through the product service life being extended and its end of life handled better because the manufacturer, which is still in control of these aspects has better knowledge than the typical customer. Additionally possible performance data is accessible for the manufacturer which will benefit in future improvement and product development. (Baines et al., 2007.) Same authors recognize that a careful assessment must be done to find out in what situations sustainable benefit can be realized. The problem is recognized also by Tukker and Tischner (2017, 71-72). As found out by them an incorrectly selected type of PSS result might even be net negative (Tukker and Tischner 2017, 90-93). Depending on the need, different types of PSS have been. According to Rennpferdt et al. (2022,1110-1112) most benefits of PSS can be realized if product is designed to be used as part of PSS.

(Design for) Recyclability

Design for recycling concentrates on design features that allow a cost-effective recycling process for products that have reached end of life. Improved recyclability of products reduces the amount of waste generated. (Ilgin et al., 2010) It can be considered handling similar challenges as design for disassembly, but it unlike design for recycling considers also the selection of materials with the target of producing "high quality waste" (Niessner 2022, 752). According to Saidani et al. (2024, 150-153) estimating the actual recyclability (including reuse and remanufacturing) of a product is challenging task. Previously different solutions, methods of calculation or software tools have been developed like one created by

Ferrão and Amaral (2006) to calculate the most cost-effective solution for recycling of passenger cars. Bras et al. (2024, 171) write that although multiple different methods have been developed in the academic world, their use is not common in industry. According to the authors best solution would be the integration of such features in modern CAD programs by developers as it is done with LCA analysis tools. Currently this kind of integrated solution is not available. Even though these tools are not yet available, designers can benefit from using traditional methods of design for recycling (Niessner, 2022, 754-756).

(Desing for) Re-manufacturing and Refurbishment

Even if recycling process manages to capture 100% of used raw materials of a product, some of originally created value is lost (Russell and Nasr 2019). Remanufacturing retains more of the original added value that was created when the product was manufactured, and such sustainability benefit while allowing reduced costs for the same reason. During remanufacturing worn or predictably failing components are replaced and the product is returned to a condition and performance comparable to a similar new product. The difference compared to service done by the user is that during re-manufacturing ownership of product changes during the process. (Sundin, 2020.)

Remanufacturing as a concept has some challenges. Shifts in technology, with advancements in product development may cause old products to be obsolete/outdated. Logistics, including storage do not add value and will incur costs and must also be considered from a sustainability point of view. Additionally, from the customer point of view remanufactured products may be less desirable which must be reflected in the sales price. (Sundin, 2020.) If the majority of a products environmental impact comes from energy consumption during its use, recycling may be a better option. (Souza, 2020.)

As with repairability, the needs of remanufacturing must be taken into account during the design process. All steps of the remanufacturing process should be considered. According to Sundin and Bras (2005, 917) these are inspection, cleaning, disassembly, repair, reassembly and testing. Part reusability is a key element of remanufacturing so the product should be easy to disassemble without damaging components and components should be easy to identify. (Lindkvist and Sundin, 2020; Sundin and Bras, 2005.)

The distinction between remanufacturing and refurbishment is based on the level to which the condition of the product is returned. According to Ijomah et al., (2007, 712)

remanufacturing returns the product to new-like condition. Russell and Nasr (2023, 27) also mention that remanufacturing is a standardized industrial approach while the goal of refurbishment is simply just to return the product to a fully functional condition.

(Design for) Repairability

Since a large part of the environmental impact of product comes from initial manufacturing, their sustainability can also be improved by extending their lifetime. Tied together with durability, repairability is important factor as it is simple way to extend lifetime of product. The benefit of repair compared refurbishment, and remanufacturing is that it uses fewer resources and typically reduces the need for transport. (Russell and Nasr, 2019.) The repairability of product is largely determined during the initial design phase of a product, with the selection materials, types of joining methods and overall geometries of the design. (Ruiz-Pastor and Mesa, 2023.) As mentioned by De Fazio, et al., (2021) many studies have been performed to create methods to assess the repairability of products, these studies highlight the importance of easy disassembly. Design for repairability is not alone enough, suitable instructions for the repair process are needed to help users perform repairs needed. Good instructions also improve the positive image of a product for customer which will increase customer willingness to extend the products service life. (Amend, et al., 2022.)

2.3 Review of practices in industry, goal and methodology

To get a benchmark of practices used in related industry, public information shared by other HVAC manufacturers of sustainable elements of their design processes was reviewed. This review helps to determine if some aspects were missed during the literature review and provides insight into the needs industry expects customers to have. It must be recognized that this information can be considered also as marketing material for the company and such will try to represent them in a positive light. Although the processes, research and discussions that led to these results are not available, some insight into the goals of company can be deduced. This should not be considered as competitor analysis, as it is just narrow look into current situation in the industry.

Review was performed in two stages like a literature review; first suitable material was searched and selected after which it was reviewed and summarized. Themes in the review

of practices are based on literature review and where new reoccurring themes appeared under those.

Material was searched by performing a web-search with search engine, with tracking data and cookies removed to reduce effect of those on the results. The search was performed by using the keywords: dampers AND fire OR ventilation, shut-off, kitchen hoods, galley hoods, cabin ventilation, HVAC systems, building ventilation, diffusers, plenums. Manufacturers of these products were reviewed and those with available data on their sustainability goals in product development were selected deeper inspection.

2.4 Interviews, their goal and methodology

The purpose of the interviews in this work is to collect Halton Group specific information related sustainability, information which is not other ways available. As Halton Group is divided into four different strategic business areas each with a different type of customers there are differences between customer wishes as well as the nature of the products. A secondary goal was to gather information how design teams have previously worked in the design process of new products in different SBAs.

Interviews were conducted with a preselected group of employees from different Halton Group SBAs. Interviewees were selected together with sustainability strategy chief officer and key personnel of Halton Group to get as good view as possible into the design and customer point of view of the products. A total of 16 persons were selected to be interviewed, which was deemed to be adequate to have good look into topic, while remaining manageable, considering due complicated subject interviews had to be long. Out of 16 planned ultimately 12 could be interviewed due scheduling challenges.

Selected interviewees were salespersons, designers, and upper management. Balance between these roles was kept so that information from all points of view could be collected. To better suit topic guides for different groups, two distinct groups of interviewees were formed. The first group consists of employees who have firsthand experience of with the products of SBU, preferably design experience. Design features relevant to sustainability are discussed with this group, with a goal to find out already tested solutions and future

possibilities. Second group consist of management who have clearer idea of future goals of the company itself. Topic guides used in interview are in Appendix 3.

Interviews were held as open and unstructured. While this method is considered challenging it was selected to reduce risk of guiding the interview in a direction where important knowledge could be left out (Tuomi and Sarajärvi, 2018). As in this case scope of interviews were all four Halton Group SBA's and the interviewers knowledge of special conditions of each SBA was limited, it was deemed more suitable to use unstructured interviews. Interviews were conducted for one person at a time.

Results of interviews were summarized under themes recognized in the literature review and where new reoccurring themes appeared new themes were added to the list. In some cases, answers were partially outside the scope of this work. Some of them were still included in the summarizations due their relevance to the topic overall.

3 Results

A review of industry practices and interviews was performed as described in chapters 2.3 and 2.4. The results of both are collected in the following subchapters.

3.1 Results of review of current industry practices

Appendix 1. contains a list of companies' guidelines that were initially reviewed. In this chapter elements that were deemed most relevant are listed. Descriptions of methods are summarized in a way that the basic idea of each method remains unchanged, but they are not direct quotes.

(Design for) Dematerialization

Reduction of waste in manufacturing by better planning and equipment (Climecon Climecon-Climate-Roadmap, 2025; Systemair, Group Sustainable Product Policy, n.d.b; AAON, 2025)

Reduction of material usage through design choices (Lindab, Sustainability report, 2025c, 39; Systemair, Group Sustainable Product Policy, n.d.b, 4)

(Design for) Material sourcing

Raw materials are purchased from suppliers who can provide materials with the smallest possible environmental footprint (Climecon Sustainability, 2025; Lindab, Sustainability report, 2025c, 31; Swegon, 2025; Trox, 2025, 38; Johnson Controls, 2025, 15; Keller-lufttechnik, 2025; AAON, 2025; Alfa-Laval, 2025; Temet 2025; Airflow, 2025)

Use of renewable materials should be increased (Climecon Climecon-Climate-Roadmap, 2025; Systemair, Group Sustainable Product Policy, n.d.b, 4; Johnson Controls, 2025, 16; Airflow, 2025)

Preferring local suppliers reduces emissions from material delivery (Climecon Climecon-Climate-Roadmap, 2025)

(Design for) Manufacture

Design for the minimum number of components (Systemair, Group Sustainable Product Policy, n.d.b, 4)

Reduce electricity consumption during manufacturing (Systemair, Group Sustainable Product Policy, n.d.b, 4)

(Design for) Transport and Distribution

Optimize product shape and size for transportation (Systemair, Group Sustainable Product Policy, n.d.b, 4)

(Design for) Efficiency

Reduction of energy usage through helping customers in proper selection and sizing of products (GreenHeck, 2025; Climecon Climecon-Climate-Roadmap, 2025; Lindab, n.d.a; Systemair, Group Sustainable Product Policy, n.d.b, 3)

Products are designed to reduce energy consumption (for example through connectivity, or demand control). (Climecon Climecon-Climate-Roadmap, 2025; Lindab, Sustainability report, 2025c, 25; Systemair, Group Sustainable Product Policy, n.d.b, 3; Trox, 2025, 46; Johnson Controls, 2025, 17; Belimo, 2023)

Regulation

Taking active role in associations and technical committees to drive the industry to more sustainable direction (GreenHeck, 2025; Lindab, n.d.b.; Trox, 2025, 38)

Increase public knowledge about good indoor air to drive demand for better indoor conditions (Lindab, Create healthy buildings, 2025c)

Helping the customer in sustainable choices

EPD's/LCA's for products help customers to select sustainable options (Climecon Climecon-Climate-Roadmap, 2025; Lindab, n.d.a; Systemair, Group Sustainable Product Policy, n.d.b, 3; Trox, 2025, 45)

(Design for) Product life extension

Improve product service life overall (Lindab, n.d.a; Trox, 2025, 58; Carrier, 2025, 22; Alfa-Laval, 2025)

(Design for) Recyclability

Design products to be reused, refurbished or recycled (Lindab, n.d.a; Trox, 2025, 56)

Avoid design choices that make recycling complicated (Systemair, Group Sustainable Product Policy, n.d.b, 5; Trox, 2025, 57)

(Design for) Re-manufacturing and Refurbishment

Products are returned to factory for refurbishment (Swecon, 2025)

(Design for) Upgradeability

Products are upgraded/updated on location (Swecon, 2025; Trox, 2025, 56)

(Design for) Repairability

Increase product connectivity to perform service just on time (Lindab, n.d.a)

Design for easy maintenance (Systemair, Group Sustainable Product Policy, n.d.b, 4; Trox, 2025, 57; AAON, 2025)

Supplied components, and co-operation with suppliers

Co-operation with suppliers to improve sustainability of purchased components (Climecon Climecon-Climate-Roadmap, 2025; Lindab, n.d.a; Systemair, Responsible Sourcing Policy, n.d.a, 2; Airflow, 2025)

Require suppliers to sign supplier code of conduct (Systemair, Responsible Sourcing Policy, n.d.a, 2)

Miscellaneous

Sustainable design guidelines are updated and improved and use of them is trained to employees (Climecon Climecon-Climate-Roadmap, 2025; Trox, 2025; Advanced Air, 2025; Airflow, 2025)

3.2 Results of interviews

Results of interviews are presented in this chapter. As the purpose of the work is to create a principles for the whole Halton Group answers obtained from interviews are not separated based on SBA but collected under different themes recognized in the literature review process. Results are not direct quotes, but summarized descriptions of interview answers.

(Design for) Dematerialization

At least part of the change should come from the legislative side because it is hard to implement more sustainable material choices if competitors use cheap materials (Stevens, 2025). Meredith (2025) mentions the same problem, from the stainless-steel point of view change is trivial, but with supplied components it will be more complicated. The use of renewable materials should be increased (Stevens, 2025). Local availability of low carbon steel limits options, but shipping from further away is not a sustainable option either (Kumar, 2025), (Nyman, 2025). Even though steel is easily recyclable it is still a very carbon intensive raw material and main challenge for Halton (Maran, 2025).

There is a risk of overdesign, features of product must be thought carefully if they are worth of extra materials (Stevens, 2025). Overall products should be designed so that they use as little materials as possible (Kumar, 2025).

Wastage during manufacturing should be minimized with good design, for example sizes of steel materials delivered are known. With better nesting, notable reductions in creation of scrap metal can be made (Kumar, 2025).

Product packaging should be considered as well during the design process (Nyman, 2025).

Current CAD systems allow partial LCA calculations for simple parts (Kauppinen, 2025).

(Design for) Efficiency

Even customers who otherwise see sustainability as meaning extra cost in product understand benefits of energy efficient products. For example, demand-based ventilation sells well in Belgium as customers understand it as a cost saving method (Stevens, 2025). According to Meredith (2025), the same sales argument is important also in SBA Foodservice in Americas. Kumar (2025) as well as Maran (2025) mention Halton MARVEL as a product where sustainability comes as extra for simple cost savings of customer. In product design, efficiency should be considered an important feature, with correct sizing of components energy usage can be reduced, by being able to use smaller actuators or by smaller ventilation fans (Hongisto, 2025). The aerodynamic performance of parts is big part of energy consumption, hard to improve but there would be benefits (Kauppinen, 2025).

Especially in cleanrooms, energy usage is a major part of the impact of products during their lifetime, in many cases over 85%. With cleanrooms, customers might be suspicious about improving energy efficiency through air volume savings in fear of reduced air hygiene (Saxen, 2025).

In cruise ships, HVAC is about 30% of the total energy consumption of the vessel. Shipping lines have recognized benefit of more efficient products, the “payback period” of more efficient systems has been realized to be much shorter than expected (Ahonen, 2025).

Helping the customer in sustainable choices:

Consulting engineers are under pressure from their customers in the construction business. This results them reusing designs, etc., so they are not actually tailored for the purpose or building. Halton Buildings have already one of the best solutions in form of HIT Design which as program allows customer easily select correctly sized components (Stevens, 2025).

According to Kumar (2025), customers would benefit from better availability of LCA calculation of Halton products. Challenge in this area is that Halton does not typically know exact details of the use environments of products so calculations can be performed only at a general level. It is not enough alone that Halton and Halton products are sustainable, we need to help customers to improve in that field as well (Maran, 2025; Koivisto, 2025).

It is important to communicate with customers, so they know what products Halton has to offer. On the other hand, Halton can listen customer requirements so that such products that customers wish can be offered. The benefit of this is also better suitability of products for their purpose, improving for example their efficiency. It can also extend time available for product development (Koivisto, 2025).

Regulation:

EU is pushing sustainability legislation, it would be beneficial for Halton to try to continue to be “ahead” (Stevens, 2025).

The mix of local standards is hindering development of more sustainable products as testing for localized standards increases development costs of new products. On the other hand, Halton has somewhat good situation in this with localized production in many countries (Stevens, 2025). A similar notable cost during product development is mentioned by Hongisto (2025) and Kauppinen (2025), this is especially relevant when considering products like fire dampers which require extensive testing by a 3rd party.

For Health and Cleanrooms standardization/guidelines come from different associations, in some cases new design guidelines have been blocked by lobbying. National and regional differences affect a lot, and customers are unwilling to take risks with new systems (Saxen, 2025).

In passenger vessels, the “net zero” goal is pushing shipping lines to improve energy efficiency of their ships, both old can new. Old trough upgrade projects and new builds trough increased demands for equipment performance (Ahonen, 2025).

In some cases, even if regulation does not advance in the field of sustainability, customers may still require it because they want to be ahead of the curve when there are ultimately changes in regulation (Maran, 2025).

Local production

The sustainability of a product can also be improved by cutting delivery distances, which Halton already excels at with its manufacturing facilities located around the world. Customer of course thinks this also through short delivery/shipping times (Stevens, 2025).

(Design for) Modularity

Customers may require customized solutions, modular products help with this without needing multiple unique product lines (Stevens, 2025), on the other hand same reason makes it hard to modularize products further, as fit for all solutions are hard to create (Kumar, 2025). Components should be standardized over different SBA’s where possible (Stevens, 2025).

On the other hand, Meredith (2025) points out that it does not make sense to globally restrict components used as local testing and certification may affect available components.

Modularity between products should be improved to reduce the number of different components to be manufactured. This could also improve material usage by allowing better cutting planning in the factory. When designing a new product, it should be checked if some component already made could be used. Currently, there are multiple different components that could be merged into one (Hongisto, 2025). This could also reduce product lead times and help during service/repair and recycling (Kumar, 2025).

(Design for) Product life extension

Customers are wishing for products that are high quality and last a long time, if a products lifetime is long, it may be more expensive than the cheaper options. Halton has proved to be able to deliver long lasting products, in some cases customers are willing to continue using 20-year-old equipment (Stevens, 2025). An increase in product warranty time would show customers that Halton is also trusts performance of its products (Stevens, 2025). In some of current projects even 60-year lifespans are required by customers (Maran, 2025).

(Design for) Product Service Systems (PSS)

Health and Cleanrooms have yearly service deals with customers, additionally auditing services to design plans for upgrades of cleanrooms (Saxen, 2025). On SBA MEI importance of service will be increased in future, different ways of providing better customer support have been planned (Ahonen, 2025; Maran, 2025).

(Design for) Recyclability:

The challenge for recyclability is large number of individual components in hoods, some products have components with hazardous materials (Hongisto, 2025). More thorough recycling information could be included with products, as well as designs for easier dismantling (Kumar, 2025). Some products have extremely long service lives, for those it is hard to estimate what kind of recyclability requirements there will be in place when they reach end of life. A challenge in the recycling process is that Halton does not have control over it. Many ships are recycled in Southeast Asia with little regulation (Maran, 2025).

(Design for) Repairability

In Halton Buildings SBA, the service and spare parts sector is not important, very few spare parts are sold (Stevens, 2025). On the other hand, Health and Cleanrooms have service deals (Stevens, 2025).

Repairability needs to be carefully considered during product design and the components/assemblies supplied as spare parts decided. With good design for repairability and serviceability a products lifetime can be extended (Hongisto, 2025).

In SBA Foodservice Americas service deals with customers are seen as future possibilities (Meredith, 2025). From service point of view, products could be more simplified to make service/repair work easier. Current designs sometimes add features at the cost of serviceability It is important to have communication between current service teams and product design (Meredith, 2025).

(Design for) Upgradeability

Products should be designed so that they can be upgraded if there are advancements in technology. In passenger vessels such upgrades have been made to kitchen hoods and cabin units with more advanced control units that control ventilation based on need. (Ahonen, 2025) Previously ability of products to be upgraded was not considered, but it is something should be concentrated on in the future. (Maran, 2025), (Kauppinen, 2025). Upgradeability usually goes hand in hand also with recyclability. (Maran, 2025). Challenge with upgrades is that non-Halton supplied systems might hinder performance of the advanced control systems designed to reduce energy consumption (Kauppinen, 2025).

Miscellaneous

Sustainability is seen as important way to separate from competitors as not everyone has yet taken it as seriously, though in some markets it is not currently competitive advantage through customers considering price/performance most important (Stevens, 2025).

Halton should be clear and easy to understand when advertising sustainability benefits to not be mixed with greenwashing and promises without any basis (Stevens, 2025). Products should have robust test data of them (Stevens, 2025).

The price of a product must be kept in mind, even if product is very good from sustainability point of view, if it is too expensive the net result in market will be negative if customer goes for cheaper (less sustainable) option from competitor (Stevens, 2025). Customers with low importance on product sustainability are also noted by Meredith (2025). With SBA Health

and cleanrooms product performance is more important marker than system cost (Saxen, 2025).

Sustainability and design requirements depend on the nature of the project. Company headquarters require different performance than kitchen of fast-food restaurant, also the location matters a lot. For some customers regulation is the only thing that pushes them to more sustainable options (Meredith, 2025).

Mindset for the correct way of thinking is an important challenge to make product design sustainable, for this to happen guidelines are needed from materials to components used (Meredith, 2025). Some of current designs are very old and were designed with little or no regard for sustainability points of view (Hongisto, 2025). It needs to be confirmed that product has a customer base/need before design work is done (Hongisto, 2025). Adequate resources needed for design work as well as competitor analysis, even on a design feature level (Hongisto, 2025). It is important to stay ahead in the product development curve (Maran, 2025).

Training is needed to help employees to understand sustainability better. It is important that the topic is understood also on a large scale and through the product lifecycle (Nyman, 2025).

Reliability and price have been the most important product features when sourcing parts from suppliers, sustainability should be considered as well but without compromising previous, spare part availability should be considered too (Nyman, 2025).

Product development is an important part of sustainability, Halton should be a leader in sustainable product development (Koivisto, 2025).

4 Discussion

Very similar themes in methods of product sustainability improvements through design were seen in the literature review, review of industry practices as well as in interviews. Even though the themes were mostly similar, there were certain differences. In the literature review, themes mostly concentrated on reduction of the effects of products at the end of life. Dematerialization and recyclability help to recover raw materials while design for modularity, re-manufacturing, refurbishment and product service systems help to extend product lifetime past the normal end of life. The review of industry practices on the other hand revealed a focus on the manufacturing stage with design for material sourcing, manufacture, transport that were not seen in the literature review. Additionally, some manufacturers gave focus on co-operation with suppliers to improve sustainability of supplied components. Interviews on the other hand recognized themes evenly from the literature review as well as from industry practices.

This difference of overall focus may be result of multiple different reasons. It may be that the manufacturing industry easily concentrates on problems that are easy to see as they are close their own operations. Problems related to use of product or disposing it are not something manufacturer faces unless they especially do research or get feedback through their service or aftersales organization. As design teams that way have better idea of these problems, there will be more numerous and more mature solutions to solve these problems compared to those appearing later in product lifecycle. It is understandable that manufacturers do not want to mention problems they have no solution ready. Meanwhile interviews performed inside Halton have more complete picture of challenges of product lifecycle. This reveal situation where possible problems and improvement areas can be freely discussed without the need of ready solution.

It must be recognized that it is also possibility that the small sample size on publications reviewed, and manufacturers has caused this difference, to confirm conclusion further research should be conducted. The following figure shows the overlap of themes between different methods used. It should be noted that there is also overlap between different themes even though this is not visible in the table.

Table 1. Sustainability themes depending on where they appeared in research

	Literature review	Industry practices	Interviews
(Desing for) Dematerialization			
(Desing for) Disassembly			
(Desing for) Efficiency			
(Desing for) Modularity			
(Desing for) Product Service Systems (PSS)			
(Desing for) Recyclability			
(Desing for) Re-manufacturing and Refurbishment			
(Desing for) Repairability / Serviceability			
(Design for) Material sourcing			
(Design for) Manufacture			
(Design for) Transport and Distribution			
(Design for) Product life extension			
(Design for) Upgradeability			
Supplied components, and co-operation with suppliers			
Helping the customer in sustainable choices			
Regulation			
Local production considered under "Transport and Distribution			

(Design for) Dematerialization as a theme was recognized in the results of all research methods. Its benefits are clear, the reduction of material used and the selection of material so that its negative impact is minimized is clear solution to the challenge of designing sustainable product. It has benefits in both during the manufacturing stage and at the end of life. Lindab and Systemair mentioned reduction of overall material volume, Lindab also together with Climecon mentioned goal of reduction of creation of waste during production. These same needs were recognized during the interviews. Additionally, it was brought up that steel being the main raw material of Halton Group products creates challenge in

dematerialization (Meredith, 2025; Maran, 2025). The use of lower CO_2 stainless steel is option if availability is good (Kumar, 2025; Nyman, 2025), but due to cost increase customers may be hesitant to purchase products made of it (Stevens, 2025). The possibility of reduction of wastage during manufacturing was noted by Kumar (2025). As can be seen dematerialization is seen as very important challenge, but solutions to solve the problem are seen limited. Easily implemented changes are limited to reduction of material, reduction of waste and changing supplier to one that can provide same raw material with a reduced impact. The problem of reducing of material usage is that while it can be performed in some products, in others it is limited by regulation, SOLAS regulation for fire dampers as an example (SOLAS, 2014, 7). Waste reduction can achieved through the standardization of components and an overall design goal of more careful product planning, taking into account the dimensions of available raw materials (Kumar, 2025). Steel with reduced environmental impact should be used if available. Due differences in material availability around the world it is not possible in all of Halton Group factory locations.

(Desing for) Disassembly was referred by Go et al. (2015) as most important factor when determining how efficiently a product can be reused or remanufactured, it was mentioned only indirectly by Systemair and Trox. In interviews it remained as a secondary mention of Kumar (2025) as benefit of modularity and feature for Design for Recyclability. It may be that disassembly is not seen as separate feature from recyclability. Due its importance in the recycling process, design for disassembly should be considered, but it could be integrated as an element for design for recyclability through design for modularity.

(Design for) Efficiency is a a clear theme when discussing product sustainability. Li et al., (2019) recognized that detailed design guidelines for designing efficient products are missing, but as Faiers et al., (2007) mentioned its benefits are seen also by customers. Multiple companies mentioned designing for energy efficiency in their publications (Climecon, 2025; Lindab, 2025, 25; Systemair, 2025, 3; Trox, 2025, 46; Johnson Controls, 2025, 17; Belimo, 2023). Repeatedly mentioned method was reduction of energy usage by correct sizing of products (GreenHeck, 2025; Climecon, 2025; Lindab, 2025; Systemair, 2025, 3). During interviews Stevens (2025) mentioned as a problem the habit of engineering

offices using calculations and drawings from previous projects to size products to save a time at the cost of suitable sizing. Another method mentioned by the industry was through connectivity or demand-controlled systems. The efficiency of the products was also recognized important in interviews. It was noted that energy efficiency works as sales argument even in markets where sustainability is seen only as extra cost (Stevens, 2025), (Meredith, 2025). There is already good experience from Halton side from these systems as improvements for ventilation systems of passenger vessels prove (Ahonen, 2025). Such Design for Efficiency can be seen essential also in future. Methods for improving energy efficiency of Halton products are improvements in aerodynamic efficiency (Kauppinen, 2025; Hongisto, 2025), demand-controlled systems as well as correctly sized and selected components. An example of benefits of correctly selected components is Belimo BF230 fire damper actuator compared to the Schischek InMax 15 and pi-safety QT.Nc. When the actuator is holding damper open BF uses 3W of power in comparison of 5W of two others (Belimo, 2025; Schischek, 2025; Pi-Safety, 2025). This seemingly insignificant difference creates notable impact, when considering that these actuators may keep dampers open all the time through 30 years of service life of ship. Large passenger vessel may have up to 1000 dampers, of which large part are fire dampers (Halton News, n.d.d). The challenge of improving the aerodynamic performance of products, is that redesigns are expensive with limited improvements. Especially with fire dampers, the costs of re-certifying are notable (Hongisto, 2025). When creating demand-controlled systems, the product must be well integrated into the complete system with other suppliers to avoid possible issues that would negate improvements (Hongisto, 2025). Because the customizable nature of Halton products customers can select some electrical components like actuators, as customers may select actuators based on other features than energy efficiency / sustainability performance design teams should work together with sales to help customers choose the most suitable option.

(Design for) Modularity is seen in publications of literature review as a tool to help to improve product sustainability, if not directly by helping to accomplish other product sustainability measures. These can be for example easier repairability, upgradeability and re-manufacturability which can lead to an extended product lifespan (Sonego et al., 2018 199-201), and recycling or disassembly at the end of the product lifecycle Go et al., (2015, 23). Interestingly modularity was not mentioned at all by the companies reviewed. On the

other hand, it was frequently mentioned in interviews, as a method of providing customers with correctly sized products (Stevens, 2025), or as a tool to reduce the number of different components manufactured and held in stock (Hongisto, 2025). Kumar (2025) also mentions its benefits during repair / service and recycling. Modularity, as well as standardization of products, could bring benefits found in the literature review to Halton products and such should be considered in the future. Meredith (2025) sees complete standardization of components as impossible due different national / regional regulations. Additionally, modularization may make heavy customization impossible, as (Kumar, 2025) notes.

(Design for) Product Service Systems provide unique possibilities by allowing total or partial control of the product to its manufacturer even after the product is in customer location. Instead of the product, customer is provided with a service, ie. something the customer would typically purchase the product to achieve. (Baines et al., 2007). As product performance is what is provided to the customer, the manufacturer can alter and adjust the product as long as the performance target is achieved. Because the manufacturer typically has better knowledge of their own products, this may result better optimization for the purpose. Product Service Systems or similar methods were not mentioned by any of the reviewed manufacturers in their sustainability materials. It may be that manufacturers that offering such services do not realize that PSS can be considered as a method of achieving better product sustainability. During interviews at Halton Group, PSS was not mentioned as a theme, but elements of Product Oriented Product Service Systems were discussed in the form of yearly service deals sold together with products (Saxen, 2025). On the SBA MEI side similar service deals are planned (Ahonen, 2025; Maran, 2025). While full PSSs would certainly provide sustainability benefits, they would require a lot of input from Halton. Considering that Halton Group does not typically deliver complete systems (except Foodservice and Health and Cleanrooms) and thus would not have complete control over HVAC systems, overall benefits would remain limited. Because mentioned reasons Product Service Systems may not be suitable for Halton Group Sustainable Design Principles. This does not mean that PSSs should not be offered where they can be offered with reasonable effort.

(Design for) Recyclability was unsurprisingly mentioned in research conducted with every method. The goal in design for recyclability is to simplify the recycling process, allow easy separation of raw materials and reduce amount of waste generated. (Ilgin et al., 2010.) Lindab and Trox mention that products should be designed so that recycling is easy, while Trox and Systemair additionally mention that design choices that make recycling more complicated should be avoided. Even though interview answers do not directly mention the importance of recyclability, it can be understood through replies. According to Hongisto (2025) large number of components in hoods makes the recycling process complicated. Maran (2025) mentions that it is hard to prepare for future recyclability requirements. Products should contain good recycling information, possibly included within product itself as recycling teams rarely have time or access to original user manuals (Kumar, 2025). Recyclability should be considered as an important aspect for Halton products in future. Recyclability should be reviewed during the product design process with suitable tools/methods. Recycling information should be prepared so that it is easily accessible for those who perform the recycling process without needing to contact Halton for instructions.

(Design for) Remanufacturing and Refurbishment can be considered as an extension of product repair as methods of extending product lifetime. During remanufacturing or refurbishment process, the product is returned to condition equal to that of a new product or its functionality is restored to the original level (Sundin, 2020). When a products usable lifetime can be extended, the ultimate recycling process can be delayed, and such added value of the original product manufacture can be retained unlike when recycling products only the value of raw materials is captured. Of the reviewed manufacturers only Swecon (2025) mentioned that products could be returned to the factory for refurbishment while it was not mentioned in Halton Group interviews at all. It is likely that the issue of transporting equipment to be refurbished or upgraded is recognized in industry as most HVAC components are large in volume. This would at the same time reduce the cost effectiveness and sustainability of the re-manufacturing/refurbishing process. Another point that may be against this design method of improving product sustainability is that HVAC equipment already has very long service lives. When a product reaches age where it would need re-manufacturing or refurbishment it is already outdated technology. The mentioned reasons show that Design for Re-manufacturing or Refurbishment should not be considered as an

element of design principles for Halton Group. Long product lifetimes (Stevens, 2025), (Maran, 2025) should direct design towards design for upgradeability.

(Design for) Repairability/Serviceability is essential for mechanical products that are designed for a long service life. If equipment must be discarded before its intended end of life if it cannot be repaired in the case of fault resources used for manufacturing whole product are lost. Russell and Nasr (2019) mention that if a product is repaired on location instead of refurbished, less resources are typically needed and need for transport is also reduced. It was noted that product repairability is in large part decided at the initial design phase (Ruiz-Pastor and Mesa 2023; Francesco et al., 2021). Because repair work is typically performed by personnel who are not very familiar with the product in addition of good repairability, adequate instructions are needed (Amend et al., 2022). Lindab (2025) mentions product connectivity as method to perform service just on time. The benefit of this would be that breakdowns could be avoided when damaged parts are replaced before they cease their function completely. Design for easy maintenance is mentioned by Systemair (2025), Trox (2025) and AAON (2025). During interviews it became clear that there are differences inside Halton Group how important the aspect of serviceability is seen. In SBA Buildings spare part sales are low (Stevens, 2025), while in SBA Foodservice and SBA MEI they are notable part of business. It is likely that the operation environment and expected product lifetimes create this difference. Overall Design for Repairability / Serviceability should be an important consideration in design principles, as sustainability benefits are clear. Still as Hongisto (2025) mentions it should be carefully considered what components/assemblies are supplied as spare parts as it is not reasonable to stock and sell every individual component.

(Desing for) Material sourcing as a theme can be seen partially under dematerialization, but as it was featured as a separate theme in industry practices review it is discussed here. Ten of manufacturers in the review mentioned that raw materials are purchased from suppliers that can provide materials with smallest possible environmental footprint. Climecon (2025) also mentions that local suppliers are to be preferred to reduce emissions from transport. They as well as Systemair (2025), Johnson Controls (2025) and Airflow (2025) also mention that use of renewable materials should be increased. This is also mentioned in interviews by Stevens (2025). As previously mentioned, local supply of steel

produced with low carbon emissions varies around the globe (Kumar, 2025; Nyman, 2025). The material sourcing side of design should be considered as part of Design for Dematerialization, but due regional differences it is hard to form general requirement.

(Design for) Manufacturibility as a theme appeared in the review of industry practices by Systemair through designing for minimum number of components and designing so that consumption of electricity during manufacturing is minimized. While these certainly provide sustainability benefits, they should be considered as part of good product design practice and thus are not necessary to mention as part of the principles.

(Design for) Transport and Distribution was brought up by Systemair (2025) in the form of optimizing product shape and size for transportation. While transportation is certainly problem with typical HVAC system components, the possibilities for improvement are limited due to duct dimensions determining product size and shape in most cases. Products should be designed for operational performance, which should not be limited by design for transport. On the other hand, product packing instructions could be created and packaging optimized to pack products to as small volume as possible, but this should be seen as a secondary target. Halton Group also has the advantage of having production around the globe, as delivery to customer can be selected from the closest location.

(Design for) Product life extension Lindab (2025), Trox (2025), Carrier (2025) and Alfa-Laval (2025) mention the goal to improve product service life. During interviews it became clear that Halton Group products are already long lasting, customers are willing to continue using over 20-year-old equipment (Stevens, 2025) and in some cases product service life is specified as long as 60 years (Maran, 2025). Stevens (2025) points out that warranties should be extended to highlight customers the long service life of products. Because it seems that the length of product service life is not a problem for Halton products it should not be targeted to extend that. In contrast, a review should be performed to determine the targeted service life and that products match it. If products are scrapped (for example when the vessel

they are installed is scrapped) while they still have usable service life left, extra material and effort used to create a longer-lasting product is lost.

Design for upgradeability is a way to extend product service life which has been recognized by Swecon (2025) and Trox (2025) based on review. Both manufacturers mention that products should be designed to allow upgrades/updates on location. With the already long service lives of HVAC components this is better method for service life extensions than remanufacturing or refurbishments as possible technology advancements that have happened since the original design of equipment can be caught up. The same is recognized in Halton as interviews show (Maran, 2025; Kauppinen, 2025). Ahonen (2025) mentions that considerable number of upgrades have already been delivered, especially for cruise ships but to foodservice projects as well. The benefit of product upgradeability is not limited to upgrades done on location, but they can help to keep older products relevant, as they make design updates easier to apply. Due to these benefits, upgradeability should be considered in product design in future, but it needs to be recognized as feature that is not needed in every product. The possibility of future upgrades must be clear.

Supplied components, and cooperation with suppliers were listed as way of improvement of sustainability by Climecon (2025), Lindab (2025), Systemair (2025) and Airflow (2025) especially in the form of working together with suppliers to improve component sustainability. While cooperation with suppliers and component manufacturers is certainly important, possibilities of affecting product design of suppliers are low. That is why the importance of this element is low, and it was not considered. Systemair (2025) also mentioned the requirement for suppliers to sign their supplier code of conduct. This was not considered to be part of work of design teams, so in this case it was left out.

Helping the customer in sustainable choices is an important aspect as incorrectly selected or sized products may negate all the improvements made in their design. It is also element of competition between manufacturers, Climecon (2025), Lindab (2025), Systemair (2025) and Trox (2025) all mention EPD’s (Environmental Product Declaration) and LCA’s (Life Cycle Assessment) as a method of helping customers to select sustainable options. Sustainability in this case is also a selling point. The importance of this aspect is seen also

in interviews, it is important to have discourse together with the customer to help them select the right products while at the same time improving products based on feedback (Koivisto, 2025; Maran, 2025). According to the Stevens (2025) HVAC consulting engineering firms are typically under tight schedules and budgets and such shortcuts are taken by reusing drawings and calculations which may result less than optimal solutions. While communication with the customer is mostly responsibility of the sales team, product design should support them with enough information to help them to guide customers to sustainable products. Halton HIT Design for SBA Building products is a good example of such, as a tool that helps customers to find information about Halton products and simulate designs.

Regulation while not directly part of the daily work of product design teams is an important part of sustainable product design. As new requirements come from legislation, different associations and customers themselves (Ahonen, 2025; Saxen, 2025; Stevens, 2025) it is important for design teams to stay up to date with current and upcoming requirements. If product is rendered obsolete due increasing demands from these directions designing replacement product may incur major costs due testing (Hongisto, 2025; Kauppinen, 2025; Stevens, 2025). As revealed by review of industry practices GreenHeck (2025), Lindab (2025) and Trox (2025) are taking active role in associations and committees to have control in these changes and to drive industry to more sustainable regulations. As can be seen it is important for design teams to keep up do date with requirements, it would also be beneficial to be present in relevant associations to at least receive up to date information about regulation/association guideline development.

There were several **Miscellaneous** themes, of which two should be highlighted. Advanced Air (2025), Airflow (2025), Climecon (2025) and Trox (2025) all mentioned the importance of keeping their Sustainable Design Guidelines up to date. It was also noted that the use of them must be trained to employees to receive maximum benefit. This was also recognized in interviews by Nyman (2025). For the future Halton Groups Sustainable Desing Principles to be utilized at fully, this requirement must be included. Stevens (2025) argued that products must have robust test data of them to separate Halton from manufacturers that use

sustainability only as a buzzword to perform greenwashing. This information could also be used to help customers to make sustainable choices, so it should be included as well.

These reviewed themes can be generally split into three categories, those that improve product sustainability at manufacturing, those that improve it during use of product and finally those that reduce product impact at the end of life. Even though the effect of these methods differs, considering answers from interviews Halton Group should not concentrate just on one area but create products that has good sustainability performance throughout their lifecycle. Also, at the beginning those methods that provide the best product sustainability improvements with the least number of changes in operation should be selected.

Based on this discussion, the themes that can be seen for Halton Group as most important are following:

- (Design for) Dematerialization, in which sustainable material sourcing is included
- (Design for) Disassembly
- (Design for) Efficiency
- (Design for) Modularity
- (Desing for) Recyclability
- (Design for) Repairability/Serviceability
- (Design for) Upgradeability
- Helping the customer in sustainable choices, requirement of robust performance test data is included in this theme
- Regulation
- Staying up to date with industry guidelines and proper implementation

5 Conclusions

The following Sustainable Design Principles are proposed for Halton Group based on research conducted in this thesis. Several discussed themes may be included under one principle, and the list was purposefully kept as short as possible to allow concentration on most important themes. They are presented in the order they are relevant during a product's service life. These should not be considered as "final" but starting point for further development together with design teams.

Understand sustainability and regulation that are related to product design

To be able to design sustainable products, product design team must understand what sustainability means related to the products they are designing and have up to date view to relevant regulations.

Without this step there is risk that sustainable design principles are not applied in the best possible way. Even if a step-by-step design guide is created for different Halton SBAs for best possible results designers need to understand the concept and goals of sustainable design. It is also important from the social side of sustainability as members of the design team may feel their work more worthwhile when they understand full purpose of design work they are doing. If product does not fulfil relevant regulation or requirements from customers (even if not required in legislation) product design process has failed.

Design for Dematerialization

This principle combines elements of dematerialization and sustainable material sourcing. The product should be designed so that its negative impact is minimized through the minimum possible material usage without degrading product performance. In addition to minimizing material usage, materials should be selected based on their sustainable performance. Purchased components should be also reviewed on these requirements.

As discovered earlier, materials determine a large part of the sustainability of a product. Due to the operating environment of many Halton Group products, it may be hard to avoid using

materials like steel, but in these cases, material usage should be minimized. In any case materials should be selected so that the most sustainable choices are preferred, for example steel manufactured with minimized carbon emissions. It must be noted that wasted materials should also be considered as material used for product, so it is important to minimize wastage during manufacturing. Similar thinking should be applied also to components purchased from suppliers, though choices in these may be limited. Supplier reliability must be considered, as well as well as product performance and reliability. If the availability of a component is suddenly limited or ended and there is need for redesign or testing possible sustainable benefits are quickly lost. The same thing happens if repeated component replacements are needed due low reliability.

Design for efficiency

Product should be designed for the best possible efficiency while retaining its performance and functionality

Because the energy consumption of product creates most of its life-cycle emissions, efficiency must be a high priority. In Halton Group products, this can be achieved through reduced pressure losses through HVAC components or by selecting more energy efficient components. As in some cases customer is selecting components used in the product, design teams should work together with sales to help customer select components that are most suitable for the purpose.

Design for upgradeability, modularity and standardization

Design for easy upgradeability/updateability, but only if there is a realistic situation where such upgrade work would be done. Products should also be as modular as possible to reduce the number of different components, which should be as standardized as possible.

In some cases, small design changes might allow easier component changes/additions to the product to improve its performance or extend its service life later. On the other hand, unnecessary upgrade options must be avoided as these may cause extra work/material usage that never gets used. Modularity makes production and customization work easier, reducing delivery times and making easier to create product that suit customer needs. Standardization

of products reduce number of different items kept in stock and may allow reduced wastage during production.

Design for serviceability

Products should be easy to service and repair. The cost of service (from a design perspective) should be such that it promotes servicing and repair instead of replacement.

Solutions that allow easy service should be used, like non-permanent connections and easy to use electrical connectors as well as adequate service space should be reserved. The need of these should still be carefully weighed based on estimated need for service or repair during lifetime of the product. Components / assemblies that will be offered as spares must be defined during product design process.

Design for Disassembly and Recyclability

After the useable lifetime of product is over, it should be easy to disassemble and recycle. Disassembly and recycling instructions should be easily accessible.

The ease of recycling should also be considered together with the product disassembly process. Main goal of this should be the easy separation of product into its base materials / recyclable components. To help this to be performed quickly and safely, instructions should be easily accessible. This could be achieved with material markings on product components and simple markings of what parts should be separated from each other.

Provide information

Make sure that adequate datasheets, manuals, customer design tools are provided to sales and aftersales teams to help them assist customers in selecting correct products for their needs.

If suitable materials are not provided there is increased risk of customers ending with wrong, or improperly sized products. This may cause unnecessary material or energy consumption.

Datasheets and manuals need to be suitable and kept up to date so that customers can easily order spare parts when needed to avoid the need for total product replacement prematurely.

These sustainable design principles are created so that they would be appropriate to use for all Halton Group SBAs. For this reason, they are kept at a general level without going into deep detail. Even so, there may be cases where not all principles are realistically applicable, in these cases design teams can use their judgement to leave them out.

To incorporate these design principles into product design process more detailed design instructions should be prepared under each principle.

Sustainability should not be considered as a solid target, even though targets and goals are set. It should be viewed as a project for perpetual improvement. For this reason, these sustainable design principles should be seen as only as a starting point, and they should be updated when there is advancement in field. Additionally, a deeper look into different principles should be taken. Because requirement for general applicability, such deeper look was not performed during this work. As found out during literature review there are different methodologies that could be used to better apply these principles to product design. This could be done after Halton Group has selected which principles they should concentrate in their product development.

Previous research on similar topics could not be found and such comparisons to similar studies could not be made. It is likely that typically this kind of research work is performed for the internal use of companies and to prevent competitors from receiving information about internal development, works are not published. On the other hand, research for the purposes of single company lacks general applicability and may be left unpublished for that reason.

Because there is no study to compare this work to, reviewing general objectivity and reliability of this work is challenging, but methods of this work were reviewed at the beginning and were found to be suitable for the purpose, and they are typically used for similar research work.

Due to the large volume of publications about sustainability, a bit over 40 reviewed in this work is rather limited look at such wide scope. With a larger number of reviewed publications multiple sources could have used which would have improved reliability.

Searching with web search engines may cause search results to be skewed due saved personal and cookie information. To avoid this during review of industry practices Google Chrome was used in incognito mode, which, according to the Google, does not personalize search results. When reviewing industry practices, the sample size was limited to 22 manufacturers because no more with sustainability related design goals could be found, which while limited gives acceptable idea of general views in industry. On the other hand, reliability of information published by manufacturers should be reviewed sceptically, as it is picked from marketing material. It is not likely that such information is purposefully incorrect, but some information may have been left away or may be inaccurate. The aspect of greenwashing must be considered as well.

Even though principles were based on and selected with information collected from the literature review, reviews of industry practices, and interviews inside Halton Group, there is a risk for work objectivity from the personal opinions of researcher.

Despite these challenges, considering the reliability of the research, it is felt that this work gives good representation of topics that should be considered when selecting sustainable design principles as source triangulation was used.

6 Compendium

Sustainability is a rather young topic of public discussion, brought up in modern sense first time in 1950's. It became well known after the Brundtland commission report "Our common future in 1987. Since then, its importance has been steadily growing, and today businesses need to consider sustainability in all decisions and operations they make.

Halton Group is currently integrating sustainability points of views into its operations. As part of this, the sustainability of new products must be considered as well. To improve on that aspect set of "Sustainable Design Principles" is needed to guide product design teams and other opertions in their work. In this work such principles were formed based on findings of research performed.

The methodology used in work includes literature review of materials found via LUT Primo database, review of industry practices found from public materials released by companies in the HVAC industry and interviews with selected employees of company from design teams, sales, and management. Based on findings following collection of Sustainable Design Principles is proposed:

- Understand sustainability and regulation that are related to product design
- Design for Dematerialization
- Design for Efficiency
- Design for Upgradeability, Modularity and Standardization
- Design for Repairability
- Design for Disassembly and Recyclability
- Provide information

With the help of these principles, design teams should be able to determine methods to achieve the design of more sustainable products. In this research application was not considered due requirement of general applicability through Halton Group and scope

mirroring that. Thus, methods of integration of principles to daily design work were not researched or discussed.

References

AAON. 2024. Aaon impact report. Accessed 17 September 2025. Available at: https://www.aaon.com/hubfs/2024 AAON Impact Report.pdf

Advanced Air. 2025. Sustainability & Carbon Reduction. Accessed 13 September 2025. Available at: https://www.advancedair.co.uk/air-distribution-and-fire/sustainability-carbon-reduction/

Alfa Laval. n.d. Energy-efficient ventilation – what are the ecological aspects of a ventilation system? Accessed 19 September 2025. Available at: https://www.ventilation-alnor.co.uk/index/support/alnor-knowledge-base/heat-recovery/energy-efficient-ventilation-%E2%80%93-what-are-the-ecological-aspects-of-a-ventilation-system.html

Airflow Developments Limited. n.d. Airflow Developments Ltd Environmental policy. Accessed 3 September 2025. Available at: https://www.airflow.com/environmental-policy

Amend, C., Revellio, F., Tenner, I., Schaltegger, S. 2022. The Potential of Modular Product Design on Repair Behavior and User Experience – Evidence from the Smartphone Industry. Journal of cleaner production 367. Web. Available at: https://doi.org/10.1016/j.jclepro.2022.132770

Ashby, M. 2012. Materials and the Environment: Eco-Informed Material Choice. 2nd ed. Waltham, MA: Butterworth-Heinemann. Available at: https://doi.org/10.1016/C2010-0-66554-0

Atisa. 2020. ECO-SUSTAINABLE RESEARCH AND DEVELOPMENT. Accessed 13 September 2025. Available at: https://atisa.it/factory/?lang=en

Baines, T., Lightfoot, H., Evans, S., Neely, A. 2007. State-of-the-art in product-service systems., Proceedings of the Institution of Mechanical Engineers, vol. 221. 1543-1552. Available at: DOI:10.1243/09544054JEM858

Barrero, Z., Barrero, D., and Dolan, C. 2021. History of Sustainability, Transparency in ESG and the Circular Economy. New York: Business Expert Press

Baldassarre, B., Calabretta, G., Karpen, I., Bocken, N., Hultink, E. 2024. Responsible Design Thinking for Sustainable Development: Critical Literature Review, New Conceptual Framework, and Research Agenda. Journal of business ethics. 195.1 (2024): 25–46. Available at: https://link.springer.com/article/10.1007/s10551-023-05600-z

Belimo. 2023. Annual report 2023. Accessed: 13 November 2025. Available at: https://report.belimo.com/ar23/en/sustainability-at-belimo

Belimo. 2023. Belimo Technical data Sheet BF230. Accessed: 14 September 2025. Available at: https://www.yumpu.com/en/document/view/27730652/technical-data-sheet-bf230-belimo

Blizzard, J., and Leidy E. 2012. A Framework for Sustainable Whole Systems Design. Design studies 33.5, 456–479. Available at: https://doi.org/10.1016/j.destud.2012.03.001

Bogue, R. 2007. Design for Disassembly: A Critical Twenty-First Century Discipline. Assembly automation 27.4, 285–289. Available at: DOI:10.1108/01445150710827069

Bras, B., and Lootens, R. 2024. Recyclability Feedback for Part Assemblies in Computer-Aided Design Software. Technology Innovation for the Circular Economy. Ed. by Nabil Nasr. Hoboken, NJ, USA: John Wiley & Sons, Inc. 183–196. Available at: https://doi.org/10.1002/9781394214297.ch15.

Carrier. 2025. 2025 sustainability and impact report. Accessed: 20 September 2025. Available at: https://www.corporate.carrier.com/Images/2025-carrier-sir-full-report-07-25-final_tcm558-236758.pdf

Carnival Corporation. 2024. 2024 Annual Report. Accessed: 1 November 2025. Available at: https://www.carnivalcorp.com/wp-content/uploads/2024/08/Carnival-Corporation-plc-2024-Annual-Report.pdf

Climecon. n.d. Sustainability. Accessed: 13 September 2025. Available at: https://climeconair.com/en-en/responsibility/

Climecon. n.d. Climate Roadmap. Accessed: 13 September 2025. Available at: https://climeconair.com/wp-content/uploads/2023/03/Climecon-Climate-Roadmap-082025.pdf

Daikin. n.d. Decarbonization- early energy modelling. Accessed: 17 September 2025. Available at: https://www.daikinapplied.com/decarbonization/knowledge-center/decarbonization-early-energy-modeling

De Fazio, F., Bakker, C., Flipsen, B. and Balkenende, R. 2021. The Disassembly Map: A New Method to Enhance Design for Product Repairability. Journal of cleaner production 320. Available at: https://doi.org/10.1016/j.jclepro.2021.128552

Faiers, A., Cook, M., and Neame, C. 2007. Towards a Contemporary Approach for Understanding Consumer Behaviour in the Context of Domestic Energy Use. Energy policy 35.8. 4381–4390. Available at: https://doi.org/10.1016/j.enpol.2007.01.003

Ferrão, P., and Amaral, J. 2006. Design for Recycling in the Automobile Industry: New Approaches and New Tools. Journal of engineering design 17.5. 447–462. Available at: DOI:10.1080/09544820600648039

FläktGroup. n.d. ESG Policy at FläktGroup. Accessed: 13 September 2025. Available at: https://www.flaktgroup.com/globalassets/our-company/sustainability/fg_esg_policy_signed.pdf/

F2A. 2019. QSE Policy. Accessed: 12 September 2025. Available at: https://www.f2a.fr/wp-content/uploads/2020/10/Politique-QSE-F2A.pdf

Go, T., Wahab, D. and Hishamuddin, H. 2015. Multiple Generation Life-Cycles for Product Sustainability: The Way Forward. Journal of cleaner production 95. 16–29. Available at: https://doi.org/10.1016/j.jclepro.2015.02.065

Greenheck. n.d. Sustainability. Accessed: 13 September 2025. Available at: https://www.greenheck.com/sustainability

Grunwald, G., Kara, A. and Spillan, J. 2025. Relationship-Oriented Sustainability Marketing: Concepts, Challenges, and Opportunities. 1st ed. 2025. Available at: DOI:10.1007/978-3-031-88349-1_7

Halton. a n.d.a Sustainability at Halton. Accessed: 13 October 2025. Available at: https://www.halton.com/sustainability-at-halton/

Halton. n.d.b History. Accessed: 13 September 2025. Available at: https://www.halton.com/contact-us/history-of-the-company/

Halton. n.d.c Foodservice Solutions. Accessed: 19 August 2025. Available at: https://www.halton.com/app/uploads/2020/09/Halton-FS-Halton-Foodservice-Solutions-uk1108.pdf

Halton. n.d.d Odyssey of the seas is equipped with 1,000 pieces of Halton dampers. Accessed: 14 September 2025. Available at: https://www.halton.com/news/odyssey-of-the-seas-is-equipped-with-1000-pieces-of-halton-dampers/

Hummel, K., and Jobst, D. 2024. An Overview of Corporate Sustainability Reporting Legislation in the European Union. Accounting in Europe 21.3. 320–355. Available at: https://doi.org/10.1080/17449480.2024.2312145

Ijomah, W., McMahon, C., Hammond, G., Newman, S. 2007. Development of Design for Remanufacturing Guidelines to Support Sustainable Manufacturing. Robotics and computer-integrated manufacturing 23.6. 712–719. Available at: DOI:10.1016/j.rcim.2007.02.017

Ilgin, M., and Gupta, S. 2010. Environmentally Conscious Manufacturing and Product Recovery (ECMPRO): A Review of the State of the Art. Journal of environmental management 91.3. 563–591. Available at: https://doi.org/10.1016/j.jenvman.2009.09.037

International Institute for Sustainable Development. 1994. Business strategy for sustainable development: Leadership and accountability for the ’90s. Available at: https://doi.org/10.1002/bse.3280030307

Johnson Controls. 2025. 2025 sustainability report. Accessed: 16 September 2025. Available at: https://www.johnsoncontrols.com/-/media/project/jci-global/johnson-controls/us-region/united-states-johnson-controls/corporate-sustainability/reporting-and-policies/2025/hq2411001_2025-sustainability-report_gri_1_2_final.pdf

Kasulaitis, B., Babbitt, C. and Tyler, A. 2021. The Role of Consumer Preferences in Reducing Material Intensity of Electronic Products. Journal of industrial ecology 25.2. 435–447. Available at: https://doi.org/10.1111/jiec.13052

Kemp-Benedict, E. 2018. Dematerialization, Decoupling, and Productivity Change. Ecological economics 150. 204–216. Available at: https://doi.org/10.1016/j.ecolecon.2018.04.020

Keller-lufttechnik. n.d. Climate and environmental protection. Accessed: 17 September 2025. Available at: https://www.keller-lufttechnik.de/en/company/climate-and-environmental-protection

Lane, R., and Watson, M. 2012. Stewardship of Things: The Radical Potential of Product Stewardship for Re-Framing Responsibilities and Relationships to Products and Materials. Geoforum 43.6. 1254–1265. Available at: https://doi.org/10.1016/j.geoforum.2012.03.012

Larrinaga, C., and Bebbington, J. 2021. The Pre-History of Sustainability Reporting: A Constructivist Reading. Accounting, auditing & accountability journal 34.9. 162–181. Available at: https://doi.org/10.1108/AAAJ-03-2017-2872

Li, L., Yu, S., and Tao, J. 2019. Design for Energy Efficiency in Early Stages: A Top-down Method for New Product Development. Journal of cleaner production 224. 175–187. Available at: https://doi.org/10.1016/j.jclepro.2019.03.137

Lindkvist H., and Sundin, E. 2020. Supporting Design for Remanufacturing: A Framework for Implementing Information Feedback from Remanufacturing to Product Design. Journal of remanufacturing 10.1. 57–76. Available at: DOI:10.1007/s13243-019-00074-7

Lindab. n.d.a Reduce the environmental impact from our customers. Accessed: 13 September 2025. Available at: https://www.lindab.com/sustainability/for-a-better-climate/sustainability-plan/reduce-the-environmental-impact-from-our-customers/

Lindab. n.d.b Create healthy buildings. Accessed: 13 September 2025. Available at: https://www.lindab.com/sustainability/for-a-better-climate/sustainability-plan/create-healthy-buildings/

Lindab. 2024c. Lindab's annual report and Sustainability Report 2024. Accessed: 13 November 2025. Available at: https://news.cision.com/lindab/r/lindab-s-annual-report-and-sustainability-report-2024,c4132690

Lähteenmäki, P. and Puustinen, T. 2019. Ilman salaisuuksia: 50 tarinaa Haltonista. Halton Group. Available at: https://www.halton.com/app/uploads/2022/09/Ilman-Salaisuuksia-50-Tarinaa-Haltonista.pdf

Moore, J., Mascarenhas, A., Bain, J., Straus, S. 2017. Developing a Comprehensive Definition of Sustainability. Implementation science: IS 12.1. Available at: https://link.springer.com/article/10.1186/s13012-017-0637-1

Niessner, N. 2022. Recycling of Plastics. 1st ed. Munich, Germany: Hanser

Noeth, K. 2024. The ESG and Sustainability Deskbook for Business: A Guide to Policy, Regulation, and Practice. 1st ed. 2024. Berkeley, CA: Apress

Pi-safety. n.d. QT.Nc Manual Pi-safety. Accessed: 25 September 2025. Available at: https://www.pi-safety.com/files/content/pdf/BA.Nc - 0003.01.DE QT.Nc.pdf

Saidani, M., Liu, X., Huey, D., Harrison, K., Wang, P., Anisi, A., Kremer, G., Greenlee, A., Shannon, T. 2024. Calculator for Sustainable Tradeoff Optimization in Multi-Generational Product Family Development Considering Re-X Performances. Technology

Innovation for the Circular Economy. Ed. by Nabil Nasr. Hoboken, NJ, USA: John Wiley & Sons, Inc. 157–169. Available at: https://doi.org/10.1002/9781394214297.ch13

Rennpferdt, C., Schneider, J., Lachmayer, R., Krause, D. 2022. Designing PSS Fleets – Consideration of the Product Architecture. Proceedings of the Design Society. vol. 2. Cambridge: Cambridge University Press. 1109–1118. Available at: DOI:10.1017/pds.2022.113

Ruiz-Pastor, L., and Mesa J. 2023. Proposing an Integrated Indicator to Measure Product Repairability. Journal of cleaner production 395. Available at: https://doi.org/10.1016/j.jclepro.2023.136434

Russell, J., and Nasr, N. 2019. Value-Retention Processes within the Circular Economy. Remanufacturing in the Circular Economy. Ed. by Nabil Nasr. United States: John Wiley & Sons. 1–2. Available at: DOI:10.1002/9781119664383.ch1

Russell, J., and Nasr, N. 2023. Value-Retained vs. Impacts Avoided: The Differentiated Contributions of Remanufacturing, Refurbishment, Repair, and Reuse within a Circular Economy. Journal of remanufacturing 13.1. 25–51. Available at: DOI:10.1007/s13243-022-00119-4

Santillo, D. 2007. Reclaiming the Definition of Sustainability (7 Pp). Environmental science and pollution research international 14.1. 60–66. Available at: https://doi.org/10.1065/espr2007.01.375

SCHAKO. 2022. Nachhaltigkeitsbericht 2022. Accessed: 13 September 2025. Available at: https://schako.com/wp-content/uploads/SCHAKO_KG_Nachhaltigkeitsbericht_2022_DRUCK_DE.pdf

Schischek. n.d. InMax 1/4 turn actuators – size S. Accessed: 14 September 2025. Available at: https://www.schischek.com/pdf/products/InMax-S-3P_en.pdf

SFS. 2018. ISO 50001 Energianhallinta, SFS-EN ISO 50001:2018. Available at: https://sfs.fi/standardeista/tutustu-standardeihin/suositut-standardit/iso-50001-energianhallinta/

Smil, V. 2025. Making the Modern World: Materials and Dematerialization. Newark, John Wiley & Sons, Incorporated.

Sonego, M., Echeveste, M. and Debarba, H. 2018. The Role of Modularity in Sustainable Design: A Systematic Review. Journal of cleaner production 176. 196–209. Available at: https://doi.org/10.1016/j.jclepro.2017.12.106

Souza, G. 2020. Remanufacturing Business Models. Remanufacturing in the Circular Economy. Ed. by Nabil Nasr. United States: John Wiley & Sons, 2020. 1–2. Available at: DOI:10.1002/9781119664383

SOLAS. 2014. RESOLUTION MSC.365(93) (2014) AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED. Accessed: 13 November 2025. Available at: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.365(93).pdf

Spalding, M., Braestrup, A. and Refosco, A. 2021. Greening the Blue Economy: A Transdisciplinary Analysis. Sustainability in the Maritime Domain. Ed. by Skinner, J., Johansson, T., and Carpenter A. Springer International Publishing. 27–59. Available at: DOI:10.1007/978-3-030-69325-1_2

Sundin, E. 2020. The Role of Remanufacturing in a Circular Economy. Remanufacturing in the Circular Economy. Ed. by Nabil Nasr. United States: John Wiley & Sons. 1–1. Available at: https://doi.org/10.1002/9781119664383.ch2

Sundin, E., and Bras, B. 2005. Making Functional Sales Environmentally and Economically Beneficial through Product Remanufacturing. Journal of cleaner production 13.9. 913–925. Available at: https://doi.org/10.1016/j.jclepro.2004.04.006

Systemair. n.d.a Group responsible sourcing policy. Accessed: 13 September 2025. Available at: https://group.systemair.com/files/Sustainability/Systemair_Group_Responsible_Sourcing_Policy_1-0.pdf

Systemair. n.d.b Group Sustainable Product Policy. Accessed: 13 September 2025. Available at: https://group.systemair.com/files/Sustainability/Group_Sustainable_Product_Policy_V1-1.pdf

Swegon. n.d. RE:3. Accessed: 13 November 2025. Available at: https://www.swegon.com/uk/sustainability/re3/

Temet. 2025. Quality and environmental policy. Accessed: 2 September 2025. Available at: https://www.temet.com/hubfs/%5BFI%5D%20Sertifikaatit%20ja%20politiikat/POL-03-B%20(EN)%20Quality%20and%20Environmental%20Policy.pdf

TROX. 2024. Sustainability report 2024. (Accessed: 14 September 2025. Available at: https://cdn.trox.de/d5f901a4fa06d9cf/f5c3627a14d1/TROX-NHB2024-Englisch-web-locked.pdf

Tsalis, T., Malamateniou, K., Koulouriotis, D., Nikolaou, I. 2020. New Challenges for Corporate Sustainability Reporting: United Nations' 2030 Agenda for Sustainable Development and the Sustainable Development Goals. Corporate social-responsibility and environmental management 27.4. 1617–1629. Available at: https://doi.org/10.1002/csr.1910

Tuomi, J., ja Sarajärvi, A. 2018. Laadullinen tutkimus ja sisällönanalyysi. Helsinki: Kustannusosakeyhtiö Tammi

Tukker, A., Tischner, U. 2025. New Business for Old Europe: Product-Service Development, Competitiveness and Sustainability. Oxford, Taylor & Francis Group. Available at: https://doi.org/10.4324/9781351280600

United Nations. n.d. The Ten principles: UN global compact The Ten Principles UN Global Compact. Accessed: 12 September 2025. Available at: https://unglobalcompact.org/what-is-gc/mission/principles

World Commission on Environment and Development. 1984. Report of the World Commission on Environment and Development: Our Common Future. Accessed: 18 September 2025. Available at: http://un-documents.net/our-common-future.pdf

WWF. n.d. Science based targets initiative WWF. Accessed: 13 September 2025. Available at: https://wwf.panda.org/discover/our_focus/climate_and_energy_practice/what_we_do/climatebusiness/science_based_targets_initiative/

Zheng, P., Chen, C., and Wang, Z. 2021. Smart Product-Service Systems. Chantilly, Elsevier. Available at: DOI:10.1016/C2020-0-02576-4

Interviews

Ahonen, S. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 28th October 2025

Hongisto, J. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 27th October 2025

Johansson, J. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 14th November 2025

Kauppinen, T. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 29th October 2025

Koivisto, A. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 30th October 2025

Kumar, P. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 28th October 2025

Meredith, P. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 24th October 2025

Nyman, A. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 29th October 2025

Saxen, A. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 27th October 2025

Shuresh, M. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 28th October 2025

Stevens, B. 2025. Sustainable design in Halton Group [Interview] with P Räsänen, 24th October 2025

Appendix 1. List of the reviewed sustainable design guidelines

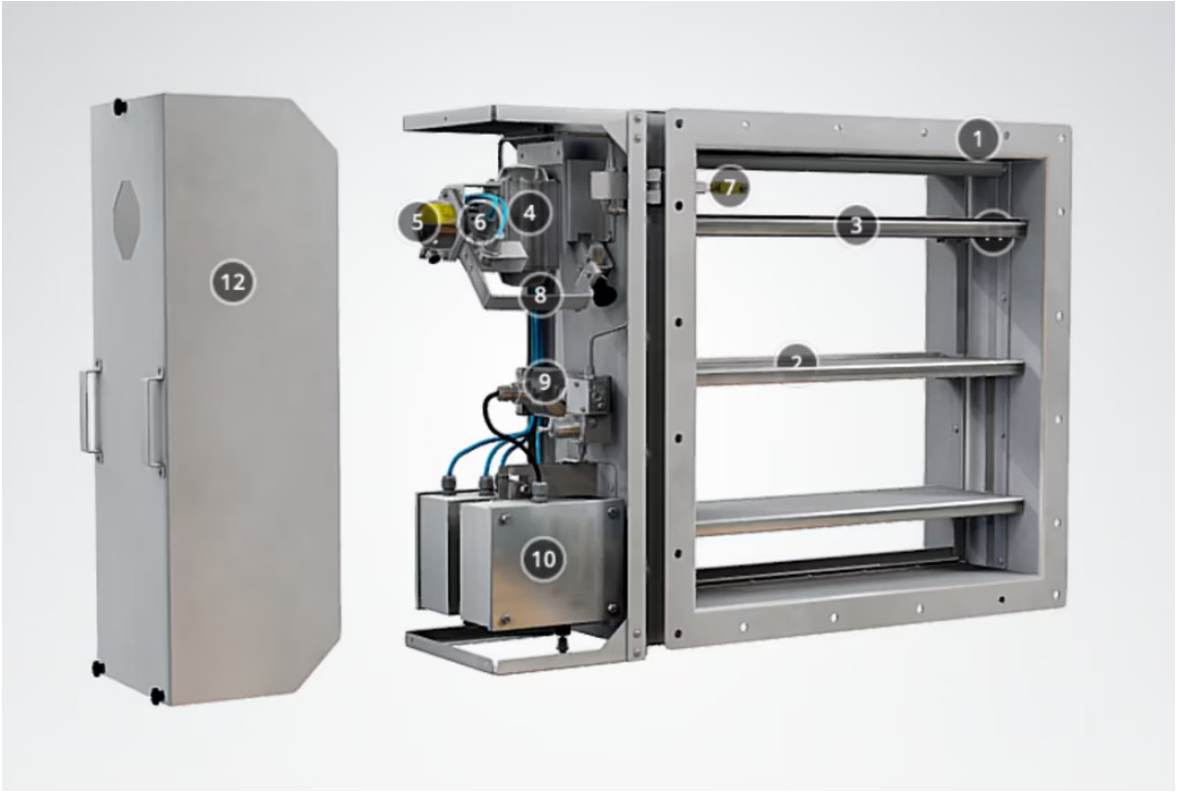
AAON
Advanced Air
Airflow
Alfa-Laval
Alnor
ATISA Spa
Belimo
Carrier
Climecon
Daikin
F2A
FläktGroup
Greenheck
Johnson Controls
Keller-lufttechnik
Lindab
Schako
Solerpalau
Systemair
Swecon
Temet
TROX

Appendix 2. Examples of Halton products

UWI, kitchen exhaust hood with water wash functionality



FDA A0(60) class marine fire damper



Droplet separator DSH

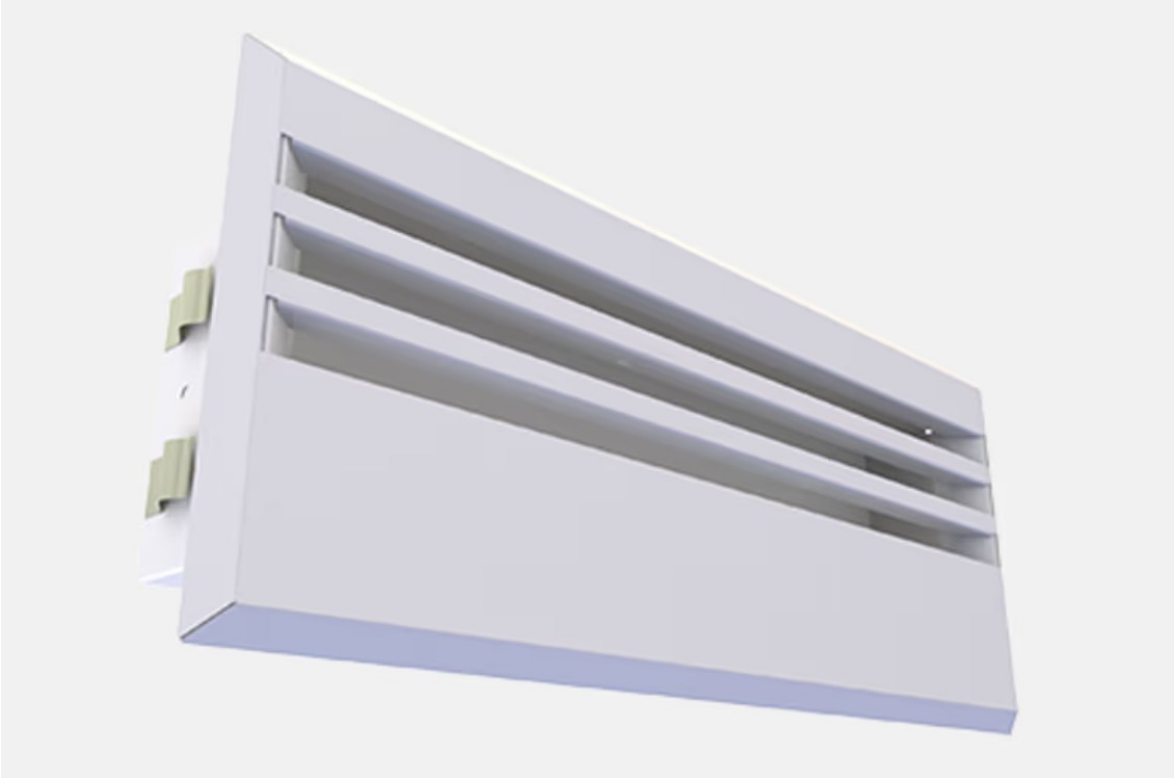


Cabin unit HMM

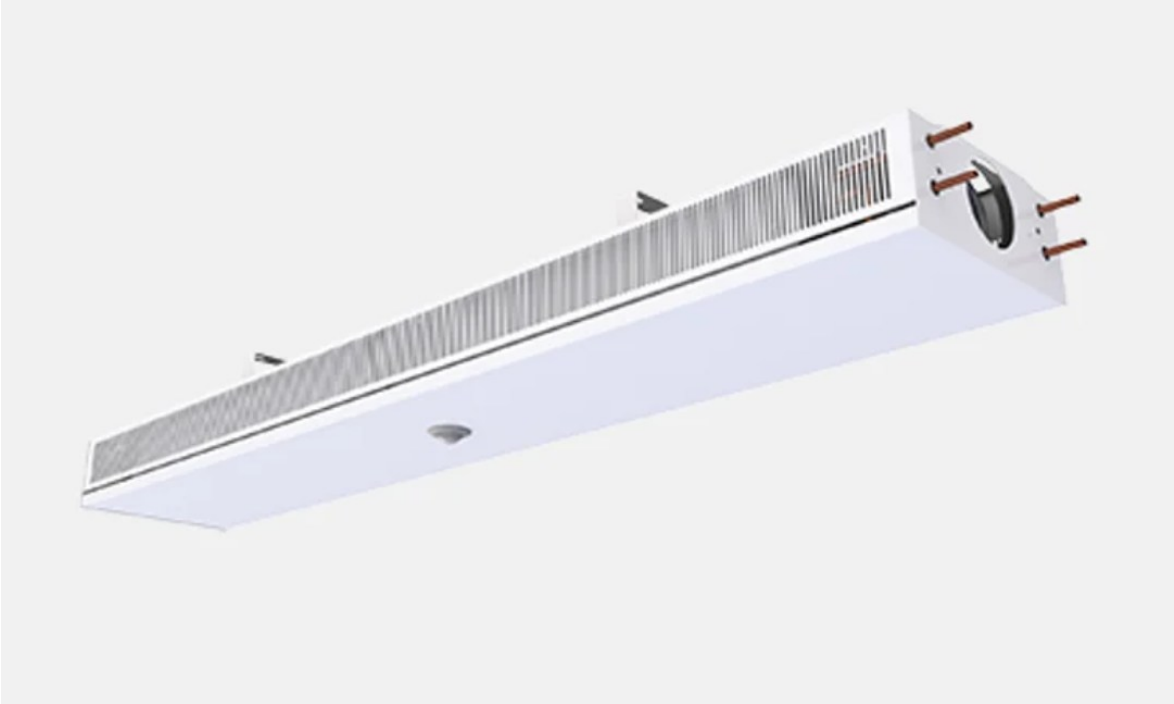


Jazz Diffuser, TLB wall diffuser





Chilled beam Rex REE



Halton, (2025) *UWI, FDA, DSH, HMM, Jazz, TLB, REE*, https://www.halton.com/products, (Retrieved: 12 September 2025).

Appendix 3. Topic guides for the interview

Technical:

- What kind of considerations there has been previously to design more sustainable product
    - o Material choices and volumes
    - o Equipment selections
    - o Manufacturing methods
    - o Product performance during its use
    - o Considerations for end of life
    - o Has product sustainability been checked previously as part of design process?
    - o Who has been responsible for that?

- What kind of possibilities / challenges can be seen in previously mentioned topics in the future?
- How Halton product groups differ from each other related to their construction?
- Should product lifecycle to be extended from current?
- How upgradeability and serviceability have been taken account in product design.

General:

- What are the legislative requirements for product sustainability?
- What are the customer requirements for product sustainability?
- How do Halton competitors market sustainability of their products?
- How could Halton use its sustainability in marketing?
    - o How to differentiate from “greenwashing”

- What is the general goal of Halton in future regarding product sustainability?
    - o Are we ready to redesign / end old product lines to replace current ones?
    - o Halton Group has extended a lot lately, has there been plans to unify products?
    - o Localization concerns?
- How should Halton compare its products to competitors related to sustainability?